Traffic Collision Analysis and Severity Prediction Using Machine Learning

Abstract - **A significant issue that poses danger to the safety of the population is the traffic accidents, which result in injuries, loss of life, and financial losses. This project examines traffic collision statistics in Toronto to determine trends which affect the severity of an accident. It is a very skewed dataset; the number of non-fatal accidents is much higher than fatal. To overcome this, resampling methods were used. Machine learning models such as Decision Tree, Random Forest and XGBoost were used to forecast the severity of accidents. Precision, recall, F1-score and ROC-AUC were used to evaluate rather than accuracy. The findings indicate that XGBoost is more accurate than other models in terms of forecasting fatal accidents. The results imply the ability of data analytics to facilitate the implementation of improved methods of traffic control and safety.**

Keywords—Traffic Collisions, Machine Learning, XGBoost, Data Imbalance, F1 Score, ROC Curve, Business Intelligence

I. INTRODUCTION

A. Traffic accidents are a major problem in the urban setting which harms human life and infrastructure. We should know the factors that lead to severity of accidents to enhance road safety. Even with access to big data, machine learning can be employed to compute the trends and forecast the results.

B. The project aims at the analysis of traffic collision data in Toronto and the development of predictive models to categorize the accidents as fatal or non-fatal. The purpose of the study is to determine trends, deal with data imbalance, and assess various machine learning models.

II. DATASET DESCRIPTION

- *The data were retrieved in Toronto Open Data and consists of the records of the traffic collisions of about 790,000.*

- *Key Features:*

  OCC_HOUR, OCC_MONTH, OCC_DOW, FATALITIES, AUTOMOBILE, PEDESTRIAN, and BICYCLE.

- *Target Variable:*

  An additional variable SEVERITY was developed:

  0 - Non-Fatal Accident

  1 - Fatal Accident

III. DATA PREPROCESSING

Preprocessing of data was done in order to prepare the data to be used in analysis and model..

Steps:.

- *Categorical variables(months,YES/NO)have been converted into numeric values.*

- *The missing values were omitted so as to maintain data quality.*

- *Developed the SEVERITY column on the basis of deaths.*

Data Imbalance:
The data was extremely skewed where there were only a few fatal cases in comparison to non-fatal cases.

Solution:

- *Balancing the dataset by applied oversampling (resampling)*

- *Guaranteed models study in both classes successfully.*

IV. EXPLORATORY DATA ANALYSIS

The data on accidents were examined using EDA to determine the patterns.

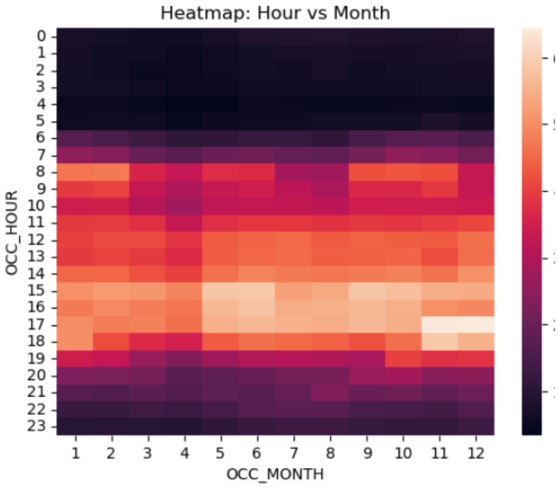
Key Observations:

- The accidents tend to happen during peak times particularly in the evening.

- There is higher frequency of accidents in some months, which are seasonal.

- The analysis of heatmaps reveals that there are concentrations of accidents at certain time intervals.

- These lessons indicate how traffic conditions influence the occurrence of accidents.



V. MACHINE LEARNING MODELS

Three machine learning models were adopted:.

A.Decision Tree

- *No complex and incomprehensible model*
- *Prone to overfitting*

B.Random Forest

- *Multiple-tree ensemble method*
- *Less overfitting and more stability*

C.XGBoost

- *Advanced boosting algorithm*
- *Deals with complicated patterns*
- *Delivers optimal performance of models*

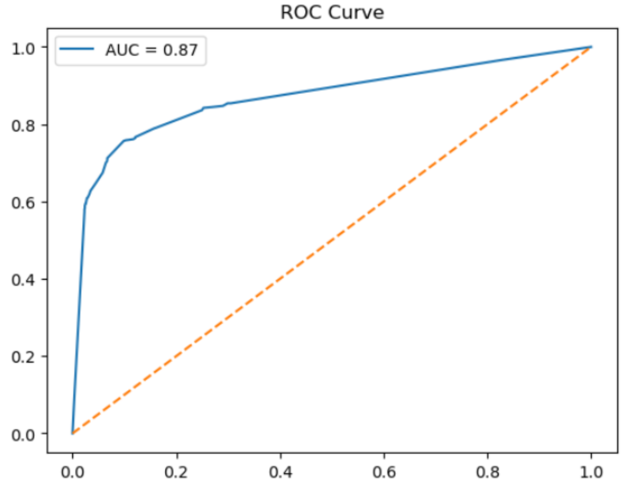
VI. MODEL EVALUATION

Accuracy was not used alone because of the imbalance of datasets.

Evaluation Metrics:

- Precision
- Recall
- F1 Score
- ROC Curve (AUC)

These measures will give a more accurate insight into the performance of the model, particularly in prediction of minority classes.



VII. RESULTS

- The performance of Decision Tree was moderate and overfitting.
- Random Forest enhanced stability and predictability.
- The best results were obtained using XGBoost that had higher F1-score and ROC-AUC.

F1 Score: 0.80

ROC-AUC: 0.87

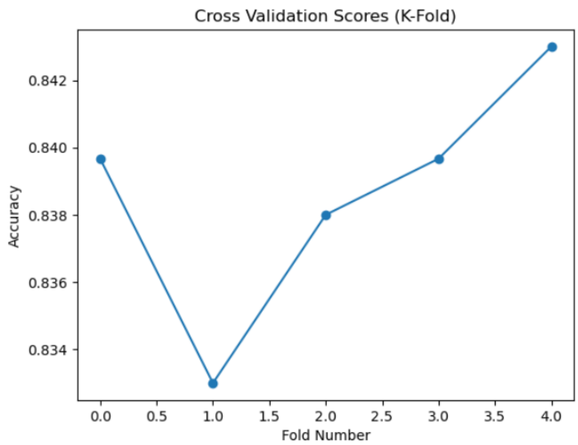
VIII. CROSS VALIDATION

It was stratified K-Fold cross-validation (5 folds).

Benefits:

- Preserves the distribution of classes in every fold.
- Gives sound performance assessment
- Reduces overfitting

Findings have validated the consistency of the XGBoost Model.



IX. Conclusion

The project evidence that machine learning is applicable in analyzing and predicting the severity of traffic accidents.

Key Points:

- The issue of data imbalance has a great impact on model performance
- Resampling enhances prediction quality and model fairness.
- XGBoost had an edge over other models because it was able to model complex relationships.

The results can help in improving traffic safety policies and decision-making.

X. FUTURE WORK

- Include more features such as weather and road conditions.
- Live prediction systems require real-time data.
- Apply deep learning models for improved accuracy.
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REFERENCES

1. **Toronto Open Data-Traffic Collisions Data.**
2. **Scikit-learn Documentation**
3. **XGBoost Documentation**